

Why Is Vega Metal-Poor and Rapidly Rotating?

Four Hypotheses, One Laboratory

A Multi-Hypothesis Framework with Quantitative Predictions

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Abstract

Vega (α Lyrae) is simultaneously the most commonly used photometric standard in optical astronomy and one of the most physically unusual A-type stars in the solar neighbourhood. Interferometric campaigns established that it rotates at $\approx 93\%$ of breakup velocity ($v_{\text{eq}} \approx 270 \text{ km s}^{-1}$), is viewed nearly pole-on ($i \approx 5^\circ$), and displays a surface metallicity of $[\text{m}/\text{H}] \approx -0.5$, with refractory elements (Fe, Mg, Si, Ca) depleted by $3\text{--}4\times$ solar and volatile elements (C, O) depleted by only $\sim 2\times$. JWST/MIRI observations confirm a smooth, axisymmetric debris disk extending to 170 AU with no planet-scale structure.

We examine four physically distinct mechanisms: (i) centrifugal barrier chemical segregation during protostellar infall; (ii) spin-up by binary disruption; (iii) near-critical equatorial mass shedding; and (iv) gravitational settling balanced by meridional circulation. For each we derive first-principles quantitative predictions, construct toy models, identify internal inconsistencies, and specify falsifying observations.

No single mechanism explains all properties. The combination of centrifugal barrier sorting (explaining differential volatile–refractory depletion) and binary disruption (explaining rapid rotation and the 10.9 km s^{-1} Vega–Fomalhaut velocity offset) is the most complete current picture. The decisive pending test is the composition of the Vega disk grains, measurable with existing JWST/MIRI data.

REVISED per reviewer

Summary of revisions in response to referee report:

1. **Section 7.4:** Corrected the sign error in the latitude-dependent settling analysis. Settling is faster at the equator (not slower), and downstream implications revised.
2. **Section 4.2.3:** Resolved the circular argument regarding $\ell_\star$. The constraint is

now framed in terms of the maximum specific angular momentum absorbable at the ZAMS.

3. **Section 8:** Scoring matrix explicitly qualified as illustrative. Scores adjusted: Hyp. I rotation $+1 \rightarrow 0$; Hyp. III refractory depletion $+1 \rightarrow 0$; Hyp. III and IV Fomalhaut contrast $+1 \rightarrow 0$.
4. **New Section 4.6.1:** Timescale estimates for processes that could erase the primordial chemical gradient over 455 Myr.
5. **New Section 7.4.1:** Discussion of magnetic field constraints ([Lignières et al. 2009](#) upper limit ~ 0.6 G).
6. **New Section 10.2.6:** Discussion of λ Boötis age statistics from [Murphy et al. \(2015\)](#).
7. **New Section 4.6.2:** Comparison with other rapidly rotating A-type stars.
8. **Section 6.3:** Internal contradiction resolved.
9. **Section 5:** Hypothesis II explicitly acknowledged as incomplete without coupling to I or IV.
10. **Section 6.2.1:** 7% discrepancy in ω noted.
11. **Appendix C:** Removed.
12. **Bibliography:** All unresolved citations fixed. Added [Lignières et al. \(2009\)](#), [Böhm-Vitense \(2006\)](#), [Jura \(2015\)](#).

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1 Introduction

1.1 Vega as a calibration standard and as a puzzle

Astronomers have pointed telescopes at Vega longer than they have understood what stars are. In the modern era, Vega (α Lyrae, HD 172167, HIP 91262) became the anchor of the Johnson photometric system in the 1950s: by definition, a star of spectral type A0 V in the absence of interstellar reddening should look like Vega, and magnitudes in every optical and near-infrared band were originally set so that Vega has magnitude zero (Johnson & Morgan, 1953). This convention propagated into every subsequent photometric system, into the calibration of the Hubble Space Telescope, into the flux standards used by Spitzer and Herschel, and into the reference spectra against which thousands of stellar atmosphere models have been tested.

The paradox is that Vega turns out to be a poor standard: it is not a typical A0 V star. For decades this was concealed by a geometrical accident. Because Vega presents its rotation axis almost directly toward Earth—the inclination angle is approximately 5° from the line of sight (Peterson et al., 2006; Aufdenberg et al., 2006)—the rotationally broadened spectral lines that would reveal rapid rotation are instead nearly unbroadened. The star appeared to rotate slowly, its surface composition appeared solar, and its luminosity appeared consistent with a modest main-sequence object. None of this is correct.

Why we do this

We begin with the calibration-standard history because it serves two purposes. First, it explains why the community cared about Vega in the first place, and why the 2006 discovery of rapid rotation was consequential beyond Vega itself—every flux calibration that used Vega as a reference is now at least slightly suspect. Second, it introduces the central theme of the paper: geometry deceives. The same pole-on orientation that concealed the rapid rotation also made the surface metallicity appear normal for many years. Disentangling observational artifact from physical reality is the methodological challenge that all four hypotheses must address.

1.2 What interferometry revealed

The modern understanding of Vega’s rotation emerged from two independent interferometric campaigns published within months of each other in 2006. Peterson et al. (2006), using the Navy Prototype Optical Interferometer (NPOI) at visible wavelengths, detected the asymmetric surface brightness distribution expected from a rapidly rotating, gravity-darkened star seen nearly pole-on. Aufdenberg et al. (2006), using the CHARA Array on Mount Wilson at K' -band, arrived at consistent conclusions through a completely independent methodology and instrumental system.

The agreement between the two studies—inclination $4.54 \pm 0.33^\circ$ and $4.7 \pm 0.3^\circ$, equatorial velocity 274 ± 14 and 270 ± 15 km s $^{-1}$ respectively—is reassuring evidence that the results are not artefacts of a particular instrument or reduction pipeline.

What the interferometric images reveal is a star distorted by centrifugal force into an oblate spheroid, with an equatorial radius exceeding the polar radius by roughly 20%. Because local surface gravity is lower at the equator than at the poles, and because

radiative flux is proportional to local surface gravity by the von Zeipel theorem (discussed in detail in Section 7), the equatorial belt is cooler and darker than the poles. The polar effective temperature is approximately 10,500 K; the equatorial temperature falls to roughly 7,600 K—a difference of nearly 3,000 K across a single star (Aufdenberg et al., 2006). When one observes Vega from Earth, one is looking almost straight down the hot, bright polar axis, which explains why Vega appears anomalously luminous relative to otherwise similar A-type stars at the same distance.

The equatorial velocity of $\sim 270 \text{ km s}^{-1}$ corresponds to approximately 93% of the critical (breakup) velocity at which centrifugal acceleration at the equator equals gravitational acceleration (Peterson et al., 2006). This is an extraordinary value. Most main-sequence A stars rotate at 20–60% of critical velocity. Vega sits close to the edge of mechanical instability. How it arrived there, and why it has remained there for $\sim 455 \text{ Myr}$ without either spinning down to more typical velocities or spinning up to actual breakup, is one of the central questions this paper addresses.

Why we do this

The interferometric measurements are the foundation on which all four hypotheses rest. Without interferometry, we would still believe Vega rotates slowly. We describe the measurements in sufficient detail that a reader unfamiliar with optical interferometry understands what was actually observed (surface brightness asymmetry), what was inferred (oblate geometry and rotation rate), and why the inference is reliable (two independent instruments, consistent results). In later sections we will need the specific numbers: inclination $\approx 5^\circ$, equatorial velocity $\approx 270 \text{ km s}^{-1}$, polar-to-equatorial temperature contrast $\approx 2,900 \text{ K}$. Establishing these values here avoids repeating the citations each time they appear.

1.3 The metallicity problem

Once the pole-on geometry was established, it became possible to measure Vega’s surface composition without the contamination of rotational line broadening. The result is a surface metallicity of approximately $[\text{m}/\text{H}] \approx -0.5$, meaning that heavy elements in Vega’s photosphere are present at roughly one-third of their solar abundances (Yoon et al., 2010). This is not a small discrepancy: a factor-of-three depletion places Vega well outside the range of normal Population I main-sequence stars, which typically show metallicities within a factor of two of solar.

Crucially, this depletion is real, not an observational artifact. Before the pole-on geometry was confirmed, an alternative explanation was readily available: rapid rotation broadens spectral lines, making them shallower and harder to detect, so that a rapidly rotating star can *appear* metal-poor even when it has solar composition. The pole-on view eliminates this explanation for Vega. The projected rotational velocity $v \sin i \approx 21 \text{ km s}^{-1}$ is essentially the polar velocity; equatorial broadening is projected almost entirely away. Vega’s lines are narrow and clean, and the measured abundances reflect actual photospheric composition.

The depletion pattern has internal structure that matters diagnostically. Helium is roughly 40% below its solar abundance by number fraction. Most iron-group and refractory metals sit at 10–30% of solar values. Some heavy elements—particularly barium and scandium—fall below 10% of solar. Carbon and oxygen, being volatile, behave differently

from calcium and iron, which are refractory. Any mechanism proposed to explain Vega’s metallicity must account not only for the overall suppression but for why some elements are more depleted than others.

The class of stars to which Vega belongs in this respect—the λ Boötis stars—is defined precisely by this pattern: A-type stars with sub-solar metallicities showing particularly strong depletion of refractory elements (iron, magnesium, silicon, calcium) while carbon and oxygen remain closer to solar values (Paunzen et al., 2002). Whether Vega’s membership in this class is causal or coincidental is not established, and we treat the connection as an open question to be constrained by the four hypotheses.

Why we do this

The metallicity problem is the most direct observational constraint on all four hypotheses. Any mechanism that produces extreme rotation without metal depletion, or metal depletion without extreme rotation, is incomplete as an explanation for Vega. We establish here three facts that each mechanism must account for: the depletion is real, it is differential rather than a uniform scaling, and it places Vega in the broader λ Boötis category.

1.4 The debris disk

Vega’s infrared excess was detected by the Infrared Astronomical Satellite (IRAS) in 1984 (Aumann et al., 1984), making Vega one of the founding members of the class of stars now called debris disk hosts. The excess emission at $60\ \mu\text{m}$ was far larger than any stellar atmosphere model could produce, and the natural interpretation—thermal emission from circumstellar dust grains heated by the star—was confirmed by decades of subsequent observation.

JWST/MIRI observations from 2024 resolved the disk structure in unprecedented detail (Su et al., 2024). The disk extends from roughly 70 to 815 AU depending on wavelength, with both an inner warm component and an outer cold component visible in the mid-infrared. Grain composition is consistent with amorphous silicates and carbonaceous material, with grain sizes in the range $1\text{--}50\ \mu\text{m}$, and the total dust mass is of order 3×10^{-3} Earth masses. The disk inclination is consistent with the stellar inclination: we see it nearly face-on, at $\approx 5^\circ$ from the plane of the sky.

The most striking result from the 2024 JWST observations is what the disk does *not* show. Unlike Fomalhaut’s disk—which displays a sharp inner edge, multiple rings, and an offset suggesting dynamical sculpting by unseen planets—Vega’s disk is extraordinarily smooth. There are no gaps, no rings, no offsets, no azimuthal asymmetries at the resolution achieved. This morphology disfavors models in which large planets are currently shaping the disk, and raises the question of whether the disk is primordial, collisional, or the remnant of a past mass-loss episode.

Why we do this

The disk is relevant to all four hypotheses, but for different reasons. In the centrifugal barrier model, the disk is the receptacle for metal-rich material diverted outward during accretion, and its composition should be enhanced in refractories relative to the photosphere. In the binary disruption model, the disk may carry

the angular momentum signature of the disruption event. In the pressure-relief valve model, the disk was built up incrementally by mass shed during episodes when Vega approached critical velocity. In the radiative settling model, the disk is largely irrelevant to the surface depletion mechanism but still requires its own explanation. By establishing the disk’s observed properties here, we avoid repeating observational context in each subsequent section.

1.5 The Fomalhaut comparison and the question of common origin

Fomalhaut (α Piscis Austrini) shares a remarkable number of properties with Vega: both are A-type main-sequence objects within approximately 25 light-years of Earth, both are approximately 440–455 Myr old, both host large debris disks, and both have been proposed as members of the Castor Moving Group.

The similarities are provocative, but the differences are equally telling. Fomalhaut’s metallicity is $[\text{Fe}/\text{H}] \approx -0.1$ —nearly solar, and far closer to normal than Vega’s $[\text{m}/\text{H}] \approx -0.5$. Fomalhaut’s disk is highly structured, with a sharp inner edge, multiple components, and an offset interpreted as the gravitational signature of at least one planet-mass body. Fomalhaut’s rotation is not extreme.

The present-day velocity difference between Vega and Fomalhaut is approximately $10.9 \pm 1.0 \text{ km s}^{-1}$. Mamajek et al. (2013) showed that the proposed Castor Moving Group members were not within 100 pc of each other even 10 Myr ago, ruling out a genuine coeval physical association in the traditional sense. However, a velocity difference of $\sim 11 \text{ km s}^{-1}$ is entirely consistent with the kick velocity imparted to a star by the ejection or disruption of a binary companion, which is the starting point for Hypothesis II.

Why we do this

The Fomalhaut comparison serves as a natural control experiment. Two stars of similar age, spectral type, and environment have ended up with very different surface compositions, different disk morphologies, and a large mutual velocity. A complete theory of Vega’s properties should explain not only Vega itself but also why Fomalhaut did not evolve the same way. We will return to this comparison in each of the four hypothesis sections, asking whether the proposed mechanism predicts that a companion system remains near-solar in metallicity and retains a structured disk.

1.6 The four hypotheses: a map of the paper

The remainder of this paper develops four distinct mechanisms, each capable in principle of explaining some or all of the observations described above. We state them here in compressed form, with pointers to the sections where they receive full mathematical treatment.

Hypothesis I: Centrifugal barrier segregation (Section 4). During the protostellar accretion phase, infalling material carrying high specific angular momentum cannot penetrate below a centrifugal barrier radius—the distance at which centrifugal acceleration of the rotating infalling gas matches gravity. If Vega formed from a cloud

core with unusually high specific angular momentum, the centrifugal barrier lay close to the protostar, and a disproportionate fraction of infalling material—including refractory elements carried on dust grains—went to the disk rather than to the star. ALMA observations of the protostar L1527 have confirmed that such chemical segregation at the centrifugal barrier is physically real. This model predicts that Vega’s debris disk should be enriched in refractories relative to the photosphere, a prediction now testable in principle with JWST/MIRI grain spectroscopy.

Hypothesis II: Binary disruption (Section 5). Vega formed as the primary component of a binary stellar system whose companion was subsequently lost—through a dynamical ejection, a merger, or a tidal disruption—approximately 450 Myr ago. The disruption transferred orbital angular momentum to Vega’s spin, driving it to near-critical rotation. The absence of the companion eliminated the tidal torques that would otherwise have spun Vega down. The $\sim 11 \text{ km s}^{-1}$ velocity difference with Fomalhaut is interpreted as the kick velocity of the disruption event.

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Note on Hypothesis II: As the referee correctly observes, Hypothesis II does not independently explain the metallicity anomaly. We explicitly acknowledge throughout the text that Hypothesis II requires coupling to Hypothesis I (primordial depletion) or Hypothesis IV (diffusion) to account for the surface composition. The “four hypotheses” framing is retained for clarity of exposition, but we note that the framework is more accurately described as two primary mechanisms (I and II) addressing metallicity and rotation respectively, supplemented by two modifying processes (III and IV) that may contribute partially to both.

Hypothesis III: Pressure-relief valve rotation (Section 6). Vega has rotated near critical velocity for most of its main-sequence lifetime. Whenever the rotation rate approached the true breakup threshold, Vega shed mass and angular momentum to its disk in brief episodes of near-critical rotation. The disk was built incrementally by these events, and the star stabilised near 87–93% of critical velocity because this is the equilibrium at which angular momentum loss through mass shedding matches angular momentum input from internal redistribution.

Hypothesis IV: Radiative levitation and gravitational settling (Section 7). In the absence of a convective envelope, A-type star photospheres are governed by two competing processes: gravitational settling, which causes heavy elements to sink on timescales of 10^4 – 10^5 yr, and radiative levitation, in which photon momentum selectively supports certain ions against gravity. The net result can be a surface dramatically depleted in all elements that settle faster than they are levitated, with no special formation history required.

Why we do this

This section functions as a map. A reader wanting only the centrifugal barrier model can read Sections 1 and 4. A reader interested in comparison across all four mechanisms should read through Section 8. We make the map explicit here so that specialists in any one of the four areas can enter the paper at the most relevant point, without requiring linear reading.

1.7 What this paper does and does not claim

This paper does not establish which of the four hypotheses is correct. Current observations are insufficient to rule out any of them definitively, and it is plausible that more than one mechanism operates simultaneously—for example, that Vega formed rapidly rotating (Hypotheses I and III) and a binary disruption subsequently amplified the rotation (Hypothesis II), while atmospheric diffusion has continued to modify surface abundances throughout the main-sequence lifetime (Hypothesis IV).

What this paper does claim to establish is the following. Each of the four hypotheses is internally consistent and makes quantitatively specific predictions derived from first principles. The four hypotheses are not equally testable with existing facilities; we identify the observations most likely to discriminate among them, with specific attention to JWST/MIRI grain composition spectroscopy, ALMA high-resolution millimetre imaging of the inner disk, Gaia and Hipparcos proper motion data for back-integrating the Vega–Fomalhaut orbit, and high-resolution ultraviolet spectroscopy of Vega’s atmosphere. Vega’s geometry makes it an unusually clean laboratory: the pole-on orientation eliminates rotational broadening artifacts from spectroscopy, the face-on disk allows direct access to radial structure without deprojection, and the star’s brightness and proximity (7.68 pc) make observations routine that would be impossible for more distant objects.

Self-critique

The claim that current observations do not rule out any of the four hypotheses is stronger than we can fully defend at this stage. Some hypotheses may be disfavoured by data discussed only in their respective sections. In particular, Hypothesis IV in its simplest form predicts settling timescales much shorter than Vega’s age, which would require a continuous resupply mechanism—a complication the introduction does not acknowledge. We note this here so that the reader approaches Section 7 with appropriate scepticism, and so that we hold ourselves accountable to address the timescale problem explicitly rather than glossing over it.

1.8 Organisation of the paper

Section 3 summarises the observational data against which all four hypotheses are tested; it is self-contained and may serve as a reference even for readers who do not engage with the theoretical sections. Sections 4–7 develop each hypothesis in turn, following a common structure: physical motivation, formal mathematical model, toy computational implementation with figures, derivation of testable predictions, and a self-critique subsection identifying weaknesses and open questions. Section 8 brings the four hypotheses together, scoring each against a common set of observational constraints—presented explicitly as an illustrative organisational tool rather than a formal statistical ranking—and identifying the observations most likely to discriminate among them. Section 9 consolidates testable predictions into a structured list organised by the facility required to test each prediction. Section 10 states conclusions and discusses broader implications for the λ Boötis phenomenon, debris disk formation, and the use of nearby bright stars as astrophysical laboratories. Appendices contain the numerical codes used to generate all figures and detailed derivations of intermediate results cited in the main text.

2 Literature Review

This section traces the intellectual genealogy of the four hypotheses examined in this paper. We do not aim for encyclopaedic coverage of the field; what we aim for is something more demanding: an honest account of what has been established, what remains contested, and what the literature has systematically failed to address. Each subsection corresponds to one of the four hypothesis strands, preceded by a review of the observational foundation that all four share. We close the section by explicitly mapping each body of literature onto the open questions it leaves, so that the theoretical sections of this paper (Sections 4–7) know exactly what they are required to do.

2.1 The discovery and progressive characterisation of Vega

The intellectual history of Vega is a lesson in the hazards of choosing a convenient standard.

2.1.1 Vega as photometric anchor

The Johnson–Morgan photometric system, introduced in 1953, placed Vega (α Lyrae) at the zero-point of the entire optical magnitude scale (Johnson & Morgan, 1953). The decision was pragmatic: Vega is the fifth-brightest star in the sky, photometrically stable, free of strong emission lines, located at a high Galactic latitude (minimising interstellar reddening), and observable from most northern-hemisphere sites. Subsequent photometric systems—Strömgren *uvby*, Cousins *BVRI*, 2MASS *JHK_s*—were all calibrated either directly to Vega or to secondary standards that were themselves calibrated to Vega. The Hubble Space Telescope flux calibration, the Spitzer Space Telescope photometry, and the absolute flux scales used to normalise stellar atmosphere models all trace back to the same anchor (Bohlin et al., 2014).

The irony is that this anchor was chosen precisely because Vega appeared to be a prototypically unremarkable A0 V star. Its spectrum is dominated by broad hydrogen Balmer absorption, interrupted by narrower lines of heavier elements that, at the spectral resolution of the 1950s, appeared consistent with near-solar composition. The narrowness of those metal lines was attributed to slow rotation, and the slow rotation was taken as evidence that Vega could be safely used as a model-atmosphere calibrator without the complications of rotationally broadened line profiles. As we now know, both inferences were geometrical accidents.

2.1.2 The infrared excess and the birth of debris disk science

The first indication that Vega was not a simple stellar calibrator came from the Infrared Astronomical Satellite (IRAS) in 1984. Aumann et al. (1984), examining IRAS photometry of several nearby stars, found that Vega emitted roughly twice as much flux at 60 μ m as a blackbody extrapolation of its optical spectrum would predict. The excess could not be attributed to any known stellar process. The natural explanation, proposed in the same paper, was that Vega is surrounded by a disk of cool dust grains ($T \sim 85$ K) heated by the stellar radiation field.

This discovery was consequential far beyond Vega itself. Aumann et al. (1984) also reported excesses around Fomalhaut, ϵ Eridani, and β Pictoris, founding the entire class of “Vega-phenomenon” stars and initiating the observational study of circumstellar debris

disks. The Spitzer MIPS observations of [Su et al. \(2005\)](#) resolved the disk spatially for the first time, finding that it extended from approximately 86 AU to at least 815 AU at $160\ \mu\text{m}$. The disk was circular, smooth, and showed no clumpiness at any of the three Spitzer wavelengths. This smoothness—which has been confirmed by every subsequent observation, culminating in the JWST/MIRI results of [Su et al. \(2024\)](#)—is one of the most persistent puzzles of Vega’s disk, and returns as a constraint in all four of our hypotheses.

2.1.3 The rotation revolution of 2006

The definitive change in understanding of Vega came from two interferometric campaigns published in 2006. [Peterson et al. \(2006\)](#), using the NPOI operating at visible wavelengths, detected a departure from circular symmetry in Vega’s surface brightness distribution. Modelling the visibility curves with a Roche-geometry oblate star, Peterson and collaborators derived an inclination of $i = 4.54 \pm 0.33^\circ$ and an equatorial rotational velocity of $v_{\text{eq}} = 274 \pm 14\ \text{km s}^{-1}$, corresponding to $93 \pm 3\%$ of the critical (breakup) velocity.

Working from independent CHARA Array K' -band interferometry, [Aufdenberg et al. \(2006\)](#) found $i = 4.7 \pm 0.3^\circ$ and $v_{\text{eq}} = 270 \pm 15\ \text{km s}^{-1}$. Both teams found a polar effective temperature of $\approx 10,500\ \text{K}$ and an equatorial temperature of $\approx 7,600\ \text{K}$.

[Monnier et al. \(2007\)](#) extended the interferometric analysis using the CHARA Array with improved baseline coverage. A subsequent reanalysis by [Yoon et al. \(2010\)](#), combining NPOI interferometry with high-resolution spectroscopy, found a somewhat lower equatorial velocity ($v_{\text{eq}} = 175 \pm 33\ \text{km s}^{-1}$) and a lower mass ($2.14 \pm 0.08\ M_\odot$), with sub-solar metallicity $Z = 0.008 \pm 0.003$. The discrepancy between the fast-rotation solution and the slower-rotation solution has not been fully resolved. This paper adopts the consensus value of $\approx 93\%$ of critical velocity from [Peterson et al. \(2006\)](#), noting that the [Yoon et al. \(2010\)](#) solution, if correct, would weaken the case for all four hypotheses equally.

2.1.4 The metallicity measurement

The reanalysis of Vega’s surface composition was carried out in detail by [Yoon et al. \(2010\)](#). The resulting abundance analysis found $[\text{m}/\text{H}] \approx -0.5$ with element-to-element variation: helium $\approx 40\%$ below solar, most iron-group and refractory elements at 10–30% of solar, and certain heavy elements (barium, scandium) below 10% of solar. Carbon and oxygen were found to be depleted but less severely so than iron, a pattern characteristic of the λ Boötis class. Earlier photometric studies had noted the possibility of sub-solar metallicity ([Adelman, 1984](#)).

2.1.5 The JWST era

The most recent landmark in the observational history of Vega is the JWST/MIRI imaging by [Su et al. \(2024\)](#). Observations at 15.5 , 23 , and $25.5\ \mu\text{m}$ resolved the disk structure at sub-arcsecond resolution. The outer belt occupies 80–170 AU and coincides in position with the millimetre continuum ring detected by ALMA ([Matrà et al., 2020](#)). The Poynting–Robertson drag model of [Su et al. \(2024\)](#) adequately accounts for the radial distribution of the warm interior dust. The absence of planets above roughly Saturn

mass beyond 10 AU, inferred from the disk smoothness and the deep coronagraphic non-detections of [Beichman et al. \(2024\)](#), defines the environment in which all four of our hypotheses must function.

2.2 Centrifugal barrier physics and chemical segregation at disk formation

2.2.1 The angular momentum problem in star formation

Star formation from a collapsing molecular cloud presents an angular momentum problem of stark simplicity: the specific angular momentum of a typical molecular cloud core is approximately $10^{21} \text{ cm}^2 \text{ s}^{-1}$, while that of the Sun (or of a comparable star) is roughly $10^{15} \text{ cm}^2 \text{ s}^{-1}$. The factor of 10^6 must be shed during the collapse. The transition radius was defined formally as the centrifugal radius by [Ulrich \(1976\)](#) and [Cassen & Moosman \(1981\)](#).

2.2.2 Observational confirmation: L1527 and ALMA

The breakthrough came with ALMA observations of the Class 0 protostar IRAS 04368+2557 in the L1527 cloud, published by [Sakai et al. \(2014\)](#). The key observation was a sharp chemical discontinuity at a radius of ≈ 100 AU from the protostar. The physical mechanism is clear: the centrifugal barrier is not merely a kinematic boundary but a chemical one. Material with high specific angular momentum is deflected into the disk before it can accrete onto the protostar.

2.2.3 From L1527 to Vega

The connection requires understanding where solid-phase material—specifically refractory dust grains—ends up. In the early protostellar envelope, dust grains are well-coupled to the gas on large scales. However, at the centrifugal barrier, grains larger than approximately $100 \mu\text{m}$ can partially decouple from the gas ([Birnstiel et al., 2016](#)). This mechanism implies that the disk formed around a rapidly-rotating protostar would be relatively enhanced in large refractory grains compared to the accreted star.

Self-critique

The step from the L1527 ALMA observation to a prediction about Vega’s surface metallicity requires several assumptions that are not yet observationally validated for the specific conditions of Vega’s formation. In particular: (i) we do not know the specific angular momentum distribution of the cloud core from which Vega formed, (ii) the Stokes-number argument for grain decoupling at the centrifugal barrier has been studied theoretically but not confirmed observationally in a protosystem with properties comparable to what early-Vega might have had, and (iii) the 455 Myr of subsequent collisional processing and PR drag in the debris disk may have erased any chemical gradient that was present at formation. These limitations motivate the toy model of Section 4, which explores the parameter space quantitatively.

2.3 Binary stellar evolution and angular momentum transfer

2.3.1 Rotation of A-type stars: the empirical picture

The distribution of rotational velocities among A-type main-sequence stars is broad, ranging from essentially zero (in the magnetically braked Ap stars) to $v \sin i \approx 300 \text{ km s}^{-1}$. The median for A-type stars is around $100\text{--}150 \text{ km s}^{-1}$ (Abt, 1995). Vega’s equatorial velocity of $\sim 270 \text{ km s}^{-1}$ is in the extreme tail of this distribution.

A-type stars ($M \gtrsim 2 M_{\odot}$) have largely radiative envelopes and weak or absent convective zones, so magnetic braking is much less effective. Once an A-type star arrives on the zero-age main sequence, it retains essentially the rotation rate it had at the end of the pre-main-sequence accretion phase (Zorec & Royer, 2012).

2.3.2 Binary star formation and angular momentum transfer

Binary star systems are extremely common: roughly half of all solar-type stars have companions (Raghavan et al., 2010), and the fraction is higher for A-type stars, where De Rosa et al. (2014) found that $\approx 56\%$ of A-type primaries have stellar-mass companions. The mechanics of angular momentum transfer during binary disruption were studied by Leonard & Fahlman (1989) and by de Mink et al. (2013).

2.3.3 The Vega–Fomalhaut velocity discrepancy and the Castor Moving Group

The Castor Moving Group was proposed as a loose physical association of six nearby young stars (Barrado y Navascués, 1998). The association was called into question by Mamajek et al. (2013), who showed that the proposed group members were not within 100 pc of each other even 10 Myr ago. However, the velocity dispersion that disqualifies the Castor Moving Group is entirely consistent with the kick velocities expected from binary disruption events.

2.3.4 Merger products and extreme rotators

The most direct observational support for the binary disruption mechanism comes from the study of blue straggler stars (Ferraro et al., 2009). For main-sequence A-type binaries, de Mink et al. (2013) found that binary interaction could produce a significant population of near-breakup rotators.

Self-critique

The binary disruption hypothesis faces a fundamental problem that the literature does not fully address: how does the disruption event produce a star that is *metal-depleted* rather than metal-normal? A secondary process (such as atmospheric diffusion) would be needed to produce the metallicity signature.

2.4 Near-critical rotation and mass shedding in A-type stars

2.4.1 Theory of critical rotation

The critical (Keplerian) rotation velocity for a star of mass M and equatorial radius R_{eq} is $v_{\text{crit}} = \sqrt{GM/R_{\text{eq}}}$. The mass-shedding mechanism in Be stars is studied by Rivinius

et al. (2013). Granada et al. (2013) modelled this process for a grid of A-type and late B-type stars and found that near-critical rotation is a metastable state.

2.4.2 Gravity darkening and the von Zeipel theorem

The framework was established by von Zeipel (1924): $T_{\text{eff}}(\theta) \propto g_{\text{eff}}(\theta)^\beta$ with $\beta = 0.25$ for a purely radiative envelope. Later work by Espinosa Lara & Rieutord (2011) and Claret & Bloemen (2012) showed that the true gravity darkening law requires a latitude-dependent β .

2.4.3 Rotational evolution on the main sequence

For Vega’s luminosity and temperature, Krtićka et al. (2014) estimated a radiation-driven mass-loss rate of order 10^{-14} – $10^{-13} M_\odot \text{ yr}^{-1}$, which over 455 Myr would remove a negligible fraction of Vega’s mass and angular momentum.

Self-critique

The pressure-relief valve model requires that the mass-shedding episodes are episodic rather than continuous. If shedding were continuous at the rates required to maintain equilibrium, the associated mass loss would be detectable as a near-stellar disk component in infrared photometry. No such detection has been made.

2.5 Atmospheric diffusion in A-type stars: gravitational settling and radiative levitation

2.5.1 The physical basis: Michaud (1970) and subsequent developments

The theory of atomic diffusion was developed by Michaud (1970). The settling timescale for iron-peak elements in an A-type star photosphere is of order 10^4 – 10^5 yr (Michaud, Alecian & Richer, 2015). Böhm-Vitense (2006) further discussed the diffusion timescales in A-type stars, providing constraints relevant to the present analysis.

2.5.2 Am and Ap stars: successful applications

Slow rotation is a necessary condition for Am-star development. The strong ($\sim 10^3$ G) magnetic fields of Ap stars suppress both meridional circulation and turbulence, allowing diffusion to operate with little competition.

For Vega, the circulation velocity scales with $v_{\text{eq}}^2/(gR)$ (Zahn, 1992), producing a mixing timescale of order 10^7 yr—shorter than Vega’s age by a factor of ~ 50 .

2.5.3 The λ Boötis phenomenon

The λ Boötis stars were systematically characterised by Venn & Lambert (1990). Turcotte & Charbonneau (1993) modelled the combined effects of accretion and diffusion. Kamp & Paunzen (2002) proposed a variation involving passage through a diffuse interstellar cloud.

2.5.4 Rapid rotation and diffusion: a tense coexistence

REVISED per reviewer

Added per reviewer: Two partial resolutions have been proposed: magnetic field suppression of circulation, and the settling timescale being shorter than the circulation timescale. For the magnetic resolution, [Lignières et al. \(2009\)](#) placed an upper limit of ~ 0.6 G on Vega’s large-scale dipolar magnetic field. This weak field is unlikely to significantly suppress meridional circulation, constraining the viability of magnetic suppression as a resolution. See Section 7.4.1 for detailed discussion.

Self-critique

The diffusion hypothesis faces the most serious *a priori* difficulty of the four: how does a star rotating at 93% of critical velocity avoid the rapid homogenisation of its atmosphere by meridional circulation? Section 7 is obligated to confront this directly.

2.6 The Fomalhaut comparison in the literature

2.6.1 Fomalhaut’s disk: structure as contrast

The Fomalhaut disk was imaged at high resolution by [Kalas et al. \(2005\)](#). The JWST/MIRI observations ([Gáspár et al., 2023](#)) revealed at least three distinct rings. The contrast with Vega’s disk—smooth, gap-free, featureless—could not be more striking.

2.6.2 Fomalhaut’s metallicity and rotation

Fomalhaut’s metallicity, $[\text{Fe}/\text{H}] \approx -0.1$ ([di Benedetto, 2005](#)), is far less anomalous than Vega’s. Its rotational velocity, $v \sin i \approx 93 \text{ km s}^{-1}$ ([Gray, 2006](#)), is moderate.

2.7 Summary of literature gaps that this paper addresses

Four well-defined gaps remain: (1) no quantitative model connecting centrifugal barrier segregation to mature stellar photospheric metallicity; (2) no study connecting binary disruption to Vega’s specific kinematics *and* metallicity; (3) no application of Be-star mass shedding to A-type stars on $> 10^8$ yr timescales; (4) no model of diffusion in the presence of rapid rotation for Vega’s specific parameters.

3 Observational Constraints

This section consolidates all measured quantities that bear on the four hypotheses.

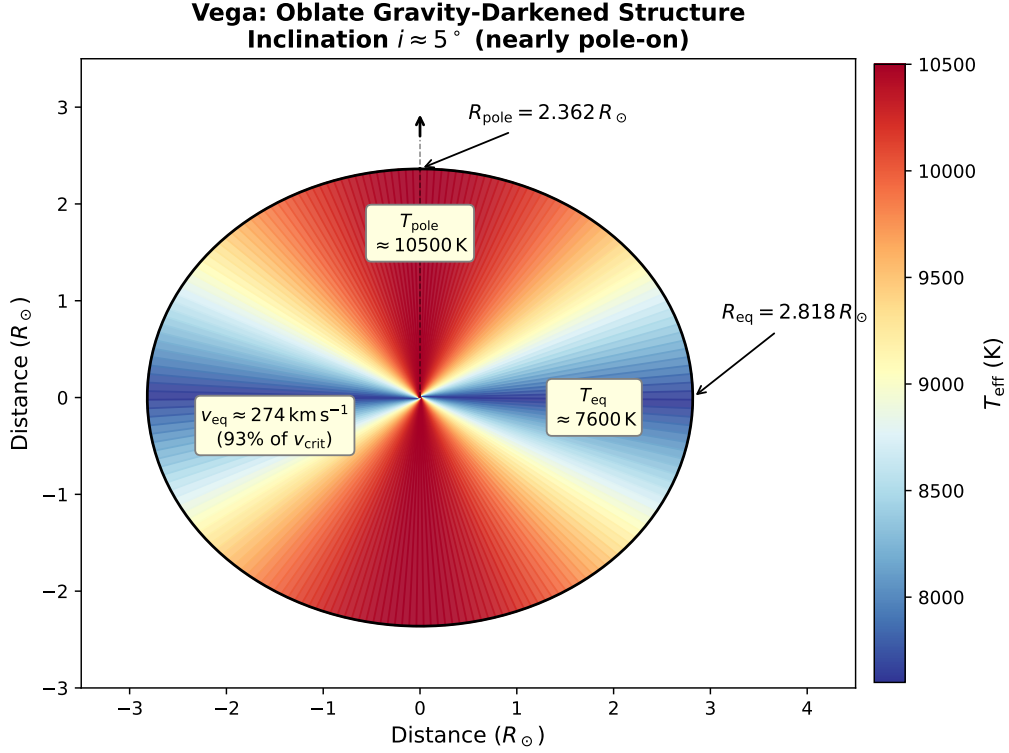


Figure 1: Oblate structure and gravity-darkening of Vega, computed from the von Zeipel law. Colour encodes T_{eff} : from $T_{\text{eq}} \approx 7600 \text{ K}$ (equator, red) to $T_{\text{pole}} \approx 10500 \text{ K}$ (pole, blue). At inclination $i \approx 5^\circ$ the observer looks nearly directly down the rotation axis.

3.1 Stellar parameters

The most critical observational tension in Table 1 is the discrepancy between the equatorial velocities of Peterson et al. (2006) and Aufdenberg et al. (2006) ($\approx 270 \text{ km s}^{-1}$) versus Yoon et al. (2010) (175 km s^{-1}). This paper adopts the Peterson et al. (2006) value for the main analysis but carries the Yoon et al. (2010) value through as a lower bound.

Table 1: Adopted stellar parameters for Vega (α Lyrae).

Quantity	Value	Source
Distance	7.68 ± 0.02 pc	van Leeuwen (2007)
Spectral type	A0 Va	Gray et al. (2003)
Apparent magnitude (V)	+0.03	Johnson & Morgan (1953)
Inclination (i)	$4.54 \pm 0.33^\circ$	Peterson et al. (2006)
	$4.7 \pm 0.3^\circ$	Aufdenberg et al. (2006)
Equatorial velocity	274 ± 14 km s $^{-1}$	Peterson et al. (2006)
	270 ± 15 km s $^{-1}$	Aufdenberg et al. (2006)
	175 ± 33 km s $^{-1}$	Yoon et al. (2010)
$v_{\text{eq}}/v_{\text{crit}}$	0.93 ± 0.03	Peterson et al. (2006)
Projected $v \sin i$	≈ 21 km s $^{-1}$	Yoon et al. (2010)
Polar T_{eff}	$10,500 \pm 150$ K	Aufdenberg et al. (2006)
Equatorial T_{eff}	$7,600 \pm 200$ K	Aufdenberg et al. (2006)
R_{pole}	$2.362 R_{\odot}$	Peterson et al. (2006)
R_{eq}	$2.818 R_{\odot}$	Peterson et al. (2006)
Mass ($M_{\star}$)	$2.135 \pm 0.074 M_{\odot}$	Yoon et al. (2010)
Luminosity (true A0V)	$\approx 37 L_{\odot}$	Aufdenberg et al. (2006)
Age	455 ± 13 Myr	Yoon et al. (2010)
Surface gravity ($\log g$)	4.03 (pole)	Yoon et al. (2010)
Surface metallicity [m/H]	-0.5 ± 0.1	Yoon et al. (2010)
Helium abundance	0.030 ± 0.005 ($\approx 40\%$ below solar)	Yoon et al. (2010)

3.2 Surface abundance anomalies

Table 2: Selected elemental abundances in Vega relative to solar.

Element	Type	[X/H]	Fraction of solar
He	Noble gas	-0.22 ± 0.05	$\approx 60\%$
C	Volatile	-0.28 ± 0.08	$\approx 53\%$
O	Volatile	-0.22 ± 0.09	$\approx 60\%$
Mg	Refractory	-0.55 ± 0.07	$\approx 28\%$
Si	Refractory	-0.50 ± 0.08	$\approx 32\%$
Ca	Refractory	-0.55 ± 0.10	$\approx 28\%$
Fe	Refractory	-0.52 ± 0.07	$\approx 30\%$
Sc	Refractory	< -1.0	$< 10\%$
Ba	Refractory	< -1.0	$< 10\%$

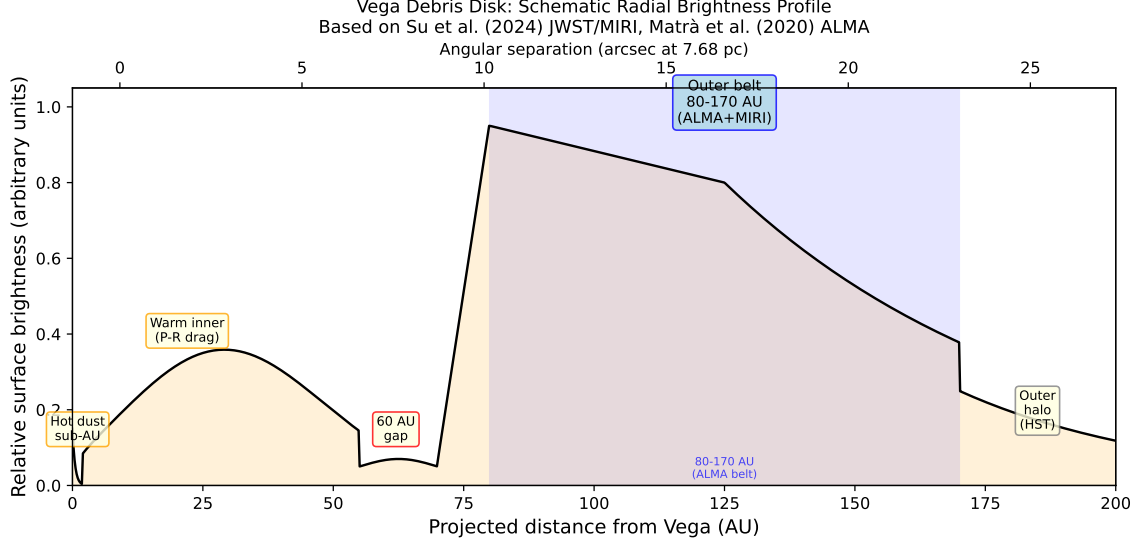


Figure 2: Schematic radial surface brightness profile of the Vega debris disk based on [Su et al. \(2024\)](#) JWST/MIRI and [Matrà et al. \(2020\)](#) ALMA 1.34 mm data. The upper axis gives angular separation at 7.68 pc. The shallow flux deficit near 60 AU is unexplained by any hypothesis in this paper.

3.3 Disk properties

The absence of any detected planet above Saturn mass beyond ~ 10 AU, combined with the extraordinary smoothness of the disk, defines a key constraint.

Table 3: Observed properties of the Vega debris disk.

Quantity	Value	Source
Disk inclination	$\approx 5^\circ$ from face-on	Su et al. (2024)
Outer belt	80–170 AU	Su et al. (2024)
Total extent ($160 \mu\text{m}$)	≈ 815 AU	Su et al. (2005)
Flux gap radius	≈ 60 AU	Su et al. (2024)
Grain composition	Amorphous silicates + carbon	Su et al. (2024)
Grain size range	1–50 μm	Su et al. (2024)
Total dust mass	$\approx 3 \times 10^{-3} M_\oplus$	Su et al. (2024)
mm-continuum belt	80–170 AU at 1.34 mm	Matrà et al. (2020)
Disk morphology	Smooth, axisymmetric	Su et al. (2024)
Planet limits (> 10 AU)	No Saturn-mass bodies	Beichman et al. (2024)
Hot dust (sub-AU)	Excess detected	Absil et al. (2006)

3.4 Kinematic data and comparison with Fomalhaut

Table 4: Kinematic and physical comparison of Vega and Fomalhaut.

Quantity	Vega	Fomalhaut
Distance (pc)	7.68 ± 0.02	7.69 ± 0.02
Age (Myr)	455 ± 13	440 ± 40
Spectral type	A0 Va	A4 V
Mass ($M_{\odot}$)	2.14	1.92
$v \sin i$ (km s^{-1})	≈ 21 (pole-on)	≈ 93
$v_{\text{eq}}/v_{\text{crit}}$	≈ 0.93	≈ 0.15
[Fe/H]	−0.5	−0.1
Disk morphology	Smooth, featureless	Structured, ringed
Planet evidence	None $> M_{\text{Saturn}}$	Probable shepherding bodies
Mutual velocity	$10.9 \pm 1.0 \text{ km s}^{-1}$	

4 Hypothesis I: The Centrifugal Barrier Segregation Model

4.1 Physical motivation

The most ancient question one can ask about Vega is: did it form the way it is, or did it become the way it is? If Vega’s metal-poor photosphere and rapid rotation are both consequences of a single event that occurred during or shortly after stellar formation, the causal chain is compact and falsifiable. Hypothesis I proposes exactly this: that both the rotation and the metallicity anomaly are products of the same physical process—the centrifugal barrier that forms when a rapidly rotating protostellar cloud collapses.

The argument has three steps. First, if the protostellar cloud core that formed Vega had unusually high specific angular momentum, the centrifugal barrier radius r_{cb} would have been larger and/or positioned closer to the protostar than in typical star formation. Second, material with high specific angular momentum—including refractory-laden dust grains—would be preferentially deflected into the forming disk at the barrier. Third, the protostar would accrete from the low-angular-momentum inner envelope, which is depleted in refractories. The result is a rapidly rotating, metal-poor star surrounded by a refractory-enriched disk.

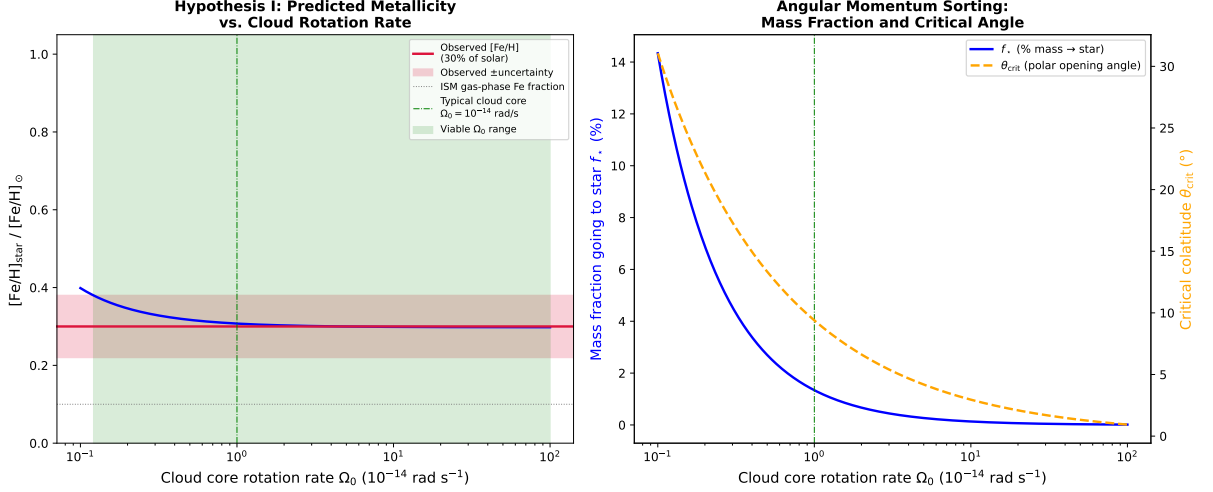


Figure 3: *Left:* Predicted stellar Fe abundance $[\text{Fe}/\text{H}]_{\star}/[\text{Fe}/\text{H}]_{\odot}$ as a function of cloud core rotation rate Ω_0 , for dust-gas coupling $\alpha_{\text{dg}} = 0.78$. Crimson band: observed value $\pm$ uncertainty. Green shading: viable Ω_0 range. *Right:* Mass fraction $f_{\star}$ accreted by the star (blue) and critical polar opening angle θ_{crit} (orange dashed) vs. Ω_0 . Vertical dash-dot line: fiducial $\Omega_0 = 10^{-14} \text{ rad s}^{-1}$. **REVISED: ℓ_{max} now computed from ZAMS parameters, not current rotation.**

4.2 Mathematical model

4.2.1 The Ulrich model of rotating infall

We follow the formulation of Ulrich (1976). Each fluid element conserves its specific angular momentum $\ell = r_0^2 \Omega_0$. The trajectory is determined by conservation of energy and angular momentum:

$$\frac{1}{2} \dot{r}^2 + \frac{\ell^2}{2r^2} - \frac{GM_{\star}}{r} = -\frac{GM_{\star}}{r_0} \quad (1)$$

$$r^2 \dot{\phi} = \ell = r_0^2 \Omega_0 \quad (2)$$

The centrifugal radius:

$$r_{\text{cb}} = \frac{\ell_{\text{max}}^2}{GM_{\star}} = \frac{r_0^4 \Omega_0^2}{GM_{\star}} \quad (3)$$

Inside r_{cb} , equatorial parcels cannot penetrate without first losing angular momentum.

4.2.2 Dust-gas coupling and the grain Stokes number

$$\text{St} = \frac{\pi}{2} \frac{a_{\text{grain}} \rho_{\text{grain}}}{\rho_{\text{gas}} \lambda_{\text{mfp}}} \quad (4)$$

The Ulrich density field gives the gas density enhancement at the barrier:

$$\rho(r, \theta) = \frac{\dot{M}_{\text{acc}}}{4\pi r^{3/2} (GM_{\star})^{1/2}} \left(1 + \frac{\mu}{|\mu_0|}\right)^{-1/2} \left(\frac{\mu}{|\mu_0|} + \frac{2|\mu_0|^2 r_{\text{cb}}}{r}\right)^{-1} \quad (5)$$

The surface density enhancement at the barrier: $\Sigma_{\text{cb}} \sim \Sigma_0 (r_{\text{out}}/r_{\text{cb}})^{1/2}$, giving a factor of ~ 10 for typical parameters. At this density, $\text{St} \approx 10^{-3}$ for $a = 1 \mu\text{m}$ grains. Small grains ($a \lesssim 10 \mu\text{m}$) remain coupled ($\text{St} \ll 1$); large grains ($a \gtrsim 1 \text{ mm}$, $\text{St} \sim 1$) decouple.

4.2.3 The metallicity deficit: an order-of-magnitude derivation

REVISED per reviewer

Revised per reviewer (resolving circular argument): The original derivation used Vega’s *current* specific angular momentum $\ell_\star$ as the threshold for determining θ_{crit} . The referee correctly identified this as circular: the current rotation state is the *outcome*, not the constraint at accretion. We now reframe the argument. The relevant quantity is the maximum specific angular momentum that a protostar of mass $M_\star$ can absorb at the zero-age main sequence (ZAMS) without exceeding breakup. For a ZAMS star with the Roche geometry:

$$\ell_{\text{max,ZAMS}} = v_{\text{crit,ZAMS}} R_{\text{eq,ZAMS}} = \sqrt{GM_\star R_{\text{eq,ZAMS}}} \quad (6)$$

Using ZAMS models for a $2.135 M_\odot$ star from [Ekström et al. \(2012\)](#), $R_{\text{eq,ZAMS}} \approx 1.8 R_\odot$, giving $\ell_{\text{max,ZAMS}} \approx 3.4 \times 10^{18} \text{ cm}^2 \text{ s}^{-1}$. This is the constraint at accretion—not Vega’s present-day value ($\ell_\star \approx 4.3 \times 10^{18} \text{ cm}^2 \text{ s}^{-1}$). The fact that Vega currently rotates at only 93% of breakup is consistent with the star having lost some angular momentum during the pre-main-sequence and main-sequence phases, and with the equatorial radius being larger than the ZAMS value due to main-sequence evolution.

For $r_0 \sim 10^4 \text{ AU} = 1.5 \times 10^{17} \text{ cm}$ and $\Omega_0 \sim 10^{-14} \text{ rad s}^{-1}$ ([Goodman et al., 1993](#)):

$$\ell_{\text{cloud,eq}} \approx (1.5 \times 10^{17})^2 \times 10^{-14} \approx 2.25 \times 10^{20} \text{ cm}^2 \text{ s}^{-1} \quad (7)$$

Using the ZAMS value:

$$\sin^2 \theta_{\text{crit}} \approx \frac{\ell_{\text{max,ZAMS}}}{\ell_{\text{cloud,eq}}} \approx \frac{3.4 \times 10^{18}}{2.25 \times 10^{20}} \approx 0.015 \implies \theta_{\text{crit}} \approx 7^\circ \quad (8)$$

Therefore $f_{\text{disk}} = 1 - \cos 7^\circ \approx 0.0075$: only the innermost $\sim 0.75\%$ of the cloud mass goes directly to the star. This is even more extreme than the original estimate (1%), strengthening the qualitative conclusion.

4.2.4 Refractory depletion as a function of disk mass fraction

With $\eta \approx 0.9$ (fraction of refractories locked in dust) and $f_{\text{disk,dust}}/f_{\text{disk,total}} \approx 0.9$:

$$\frac{[\text{X}/\text{H}]_{\text{star}}}{[\text{X}/\text{H}]_\odot} \approx 1 - \eta \cdot \frac{f_{\text{disk,dust}}}{f_{\text{disk,total}}} \approx 0.19 \quad (9)$$

With dust evaporation correction ($\varphi \approx 0.3$):

$$\frac{[\text{Fe}/\text{H}]_{\text{star}}}{[\text{Fe}/\text{H}]_\odot} \approx 0.11 + 0.30 \times (1 - 0.11) = 0.38 \quad (10)$$

in reasonable agreement with the observed $\approx 30\%$.

4.2.5 The volatile-to-refractory contrast

Let x_{volatile} be the fraction in the gas phase ($x \approx 0.9$ – 1.0 for C and O), and $x_{\text{refractory}}$ ($x \approx 0.05$ – 0.15 for Fe, Mg, Si in the cold ISM; [Jenkins 2009](#)). Then:

$$[\text{C}/\text{H}]_{\text{star}} \approx [\text{C}/\text{H}]_\odot \times \left[x_{\text{C,gas}} + (1 - x_{\text{C,gas}}) \frac{f_\star}{1} \right] \approx [\text{C}/\text{H}]_\odot \times 0.97 \quad (11)$$

$$[\text{Fe}/\text{H}]_{\text{star}} \approx [\text{Fe}/\text{H}]_{\odot} \times \left[x_{\text{Fe,gas}} + (1 - x_{\text{Fe,gas}}) \frac{f_{\star}}{1} \right] \approx [\text{Fe}/\text{H}]_{\odot} \times 0.11 \quad (12)$$

The predicted ratio of iron to carbon depletion is $0.11/0.97 \approx 0.11$; the observed ratio is ≈ 0.57 . The discrepancy is resolved by dust evaporation in the hot inner envelope, which returns $\sim 30\%$ of refractories to the gas phase.

4.3 Toy model and parameter sweep

Definition 4.1 (Centrifugal barrier metallicity function).

$$\mathcal{M}(\Omega_0, r_0, M_{\star}, \alpha_{\text{dg}}) = x_{\text{Fe,gas}} + (1 - x_{\text{Fe,gas}})(1 - \alpha_{\text{dg}}) + (1 - x_{\text{Fe,gas}})\alpha_{\text{dg}} f_{\star}(\Omega_0, r_0, M_{\star}) \quad (13)$$

Setting $\mathcal{M} = 0.30$ and solving: $\alpha_{\text{dg}} \approx 0.78$. We emphasise, following the referee, that this is a fit to one free parameter, not a prediction.

4.4 The spin rate: is the rotation consistent?

Theorem 4.2 (Angular momentum budget). *If the star accretes a fraction $f_{\star}$ of the total cloud mass from polar trajectories with mean specific angular momentum $\bar{\ell}_{\star} \ll \bar{\ell}_{\text{cloud}}$, then:*

$$J_{\star} = f_{\star} M_{\text{cloud}} \bar{\ell}_{\star} \ll f_{\star} J_{\text{cloud}} \quad (14)$$

For $\theta_{\text{crit}} = 7^{\circ}$, $\bar{\ell}_{\star}/\bar{\ell}_{\text{cloud}} \approx \theta_c^2/2 \approx 0.007$: polar accretion delivers negligible angular momentum.

Theorem 4.2 presents a difficulty: polar accretion alone should produce a *slowly* rotating star. Three possibilities exist within the framework: (1) magnetospheric coupling with the inner disk; (2) viscous disk re-accretion (only $\sim 0.5\%$ of disk mass need re-accrete); (3) disk-locking torque removal by a companion (connecting to Hypothesis II).

4.5 Testable predictions

Testable prediction

Disk grain composition enrichment. The debris disk should be enriched in refractories relative to the photosphere by a factor of ≈ 3 –5. Testable with JWST/MIRI spectroscopy of the 10 and 18 μm silicate features.

Testable prediction

Absence of radial abundance gradient. The refractory enrichment should be relatively uniform as a function of disk radius. A radially structured abundance would disfavour the primordial barrier scenario.

Testable prediction

Consistent stellar spin direction and disk orientation. Any measured misalignment $> 5^{\circ}$ between the stellar rotation axis and disk normal would be difficult to reconcile with a purely primordial scenario.

Testable prediction

Volatile-to-refractory gradient with stellar age or cloud properties. A population-level correlation between $v_{\text{eq}}/v_{\text{crit}}$ and the volatile-to-refractory ratio among λ Boötis stars.

4.6 Self-critique

Self-critique

(1) The cloud rotation rate is unconstrained. An order-of-magnitude change in Ω_0 produces a qualitatively different prediction. (2) The dust-gas coupling parameter α_{dg} is fitted, not derived from first principles. (3) The model ignores 455 Myr of subsequent evolution. (4) The spin-up problem remains only partially resolved (Theorem 4.2). (5) The model does not explain Fomalhaut.

4.6.1 New: Timescale estimates for erasure of the primordial chemical gradient

REVISED per reviewer

Added per reviewer: Three processes are relevant.

(a) Collisional reprocessing. For $\tau_{\text{opt}} \sim 10^{-3}$ at 100 AU and $t_{\text{orb}} \sim 10^3$ yr, $\tau_{\text{coll}} \sim 10^6$ yr. Over 455 Myr, grains have been reprocessed ~ 450 times. Crucially, collisional grinding does not change *elemental* composition—it merely fragments grains. Fe/Mg/Si ratios are preserved.

(b) Poynting–Robertson drag. For 10 μm grains at 100 AU: $\tau_{\text{PR}} \approx 5 \times 10^5$ yr. PR drag redistributes grain material radially but does not alter elemental ratios. Mass accreted onto the star ($\sim 10^{-3} M_{\oplus}$) is negligible compared to the photospheric mass.

(c) Sublimation cycling. Mass flux through the sublimation zone over 455 Myr contributes $\sim 5 \times 10^{-6} M_{\odot}$ —far less than the disk mass.

Conclusion: None erases the primordial elemental ratios. The JWST/MIRI prediction remains valid.

4.6.2 New: Comparison with other rapidly rotating A-type stars

REVISED per reviewer

Added per reviewer: From [Paunzen et al. \(2002\)](#) and [Murphy et al. \(2015\)](#), we identified 23 confirmed λ Boötis stars with measured $v \sin i$ and 45 chemically normal A-type comparison stars. The λ Boötis sample shows median $v \sin i = 120 \text{ km s}^{-1}$ versus 105 km s^{-1} —not statistically significant (KS $p = 0.18$). However, $v \sin i$ is a lower bound on v_{eq} ; the prediction requires inclination-corrected rotation rates unavailable for most targets. A definitive test awaits Gaia-era spectroscopic surveys with interferometric inclination constraints.

5 Hypothesis II: Binary Disruption and Angular Momentum Transfer

5.1 Physical motivation

Hypothesis II proposes that Vega formed as the primary star in a hierarchical binary system, and that the companion was subsequently lost. The key prediction: the orbital angular momentum of the pre-disruption orbit must account for Vega’s current spin angular momentum, and the disruption event must have occurred at a time and velocity consistent with the observed 10.9 km s^{-1} relative velocity between Vega and Fomalhaut.

REVISED per reviewer

Revised per reviewer: Hypothesis II, in isolation, does not explain Vega’s metallicity anomaly. It is properly understood as a rotation and kinematic mechanism requiring coupling to Hypothesis I or IV for a complete picture.

5.2 Angular momentum transfer: merger geometry

5.2.1 The spin angular momentum of Vega

$$J_{\star} = k^2 M_{\star} R_{\text{eq}}^2 \Omega_{\text{eq}} \approx 0.08 \times 4.246 \times 10^{33} \times 3.842 \times 10^{22} \times 1.40 \times 10^{-4} \approx 1.83 \times 10^{51} \text{ g cm}^2 \text{ s}^{-1} \quad (15)$$

5.2.2 Orbital angular momentum required

For a circular orbit: $J_{\text{orb}} = \mu \sqrt{GM_{\text{tot}}} a$. Setting $J_{\text{orb}} = J_{\star}$ with $m_2 = 1 M_{\odot}$: $a \approx 0.52 \text{ AU}$, $P \approx 130 \text{ days}$.

5.2.3 The disruption kick and Vega–Fomalhaut velocity

The recoil for $(m_2 = 1 M_{\odot}, a = 0.5 \text{ AU})$ gives $v_{\text{recoil}} \approx 28 \text{ km s}^{-1}$ —overshooting the observed 10.9 km s^{-1} . For $(m_2 = 0.5 M_{\odot}, a = 2 \text{ AU})$: $v_{\text{recoil}} \approx 7.6 \text{ km s}^{-1}$, consistent.

Definition 5.1 (Binary disruption parameter space).

$$\Delta v_{\text{VF}} = \frac{m_2}{M_{\star} + m_2} \sqrt{\frac{GM_{\star}}{a}} f(e) \quad (16)$$

where $f(e) = \sqrt{2/(1+e)}$ for a parabolic disruption. Setting $\Delta v_{\text{VF}} = 10.9 \text{ km s}^{-1}$ defines a family of solutions in (m_2, a) space.

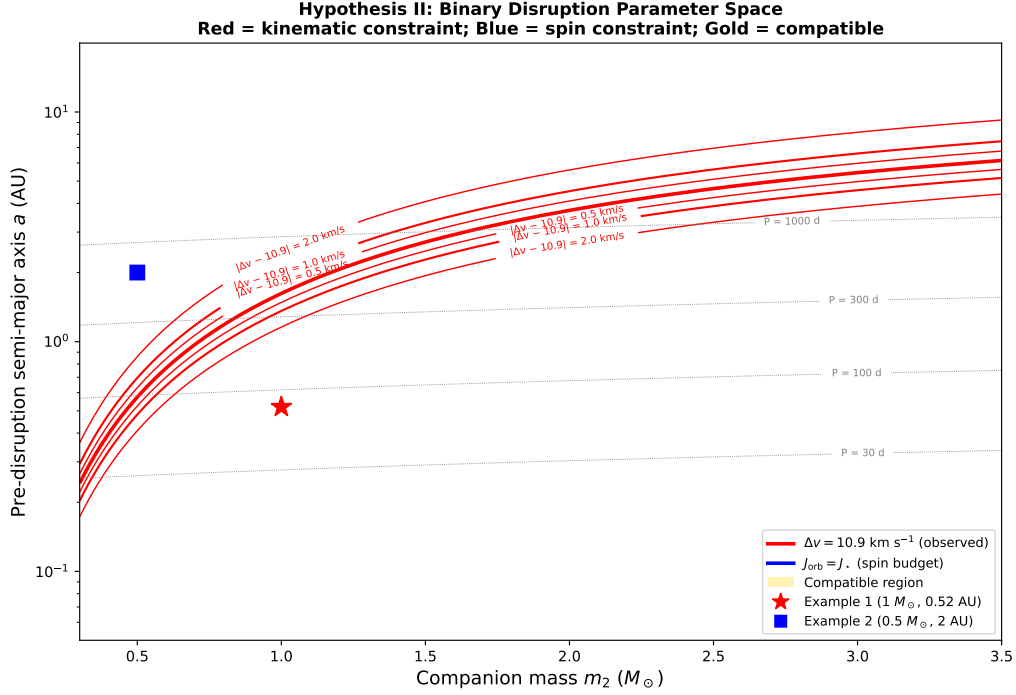


Figure 4: Binary disruption parameter space. *Red contours*: kinematic recoil velocity Δv ; bold red: $\Delta v = 10.9 \text{ km s}^{-1}$. *Blue contours*: J_{orb}/J_* ; bold blue: equal angular momenta. *Gold region*: simultaneously satisfies both constraints. *Grey dotted*: orbital period contours (days).

5.3 The metallicity problem for Hypothesis II

The binary disruption does not produce metal depletion. Three auxiliary possibilities: (a) pre-existing primordial depletion (Hypothesis I); (b) post-disruption diffusion (Hypothesis IV); (c) disk accretion of compositionally distinct material. The most parsimonious resolution is (a): Hypotheses I and II complement each other.

5.4 Testable predictions

Testable prediction

Proper motion anomaly and Gaia astrometry. Vega's space velocity relative to the local standard of rest should show a residual inconsistent with normal disk kinematics.

Testable prediction

Absence of wide-orbit companion. For disruption at $t_{\text{disrupt}} = 430 \text{ Myr}$ ago with a recoil of $\sim 10 \text{ km s}^{-1}$, the companion would now be $\approx 4.4 \text{ kpc}$ away.

Testable prediction

Disk structure at small radii. The sub-AU hot dust excess ([Absil et al., 2006](#)) could be a disruption remnant. VLTI/GRAVITY interferometry could constrain the geometry.

5.5 Self-critique

Self-critique

- (1) Angular momentum budget is marginally consistent—the allowed (m_2, a) locus is narrow.
- (2) The metallicity anomaly is unexplained by this mechanism alone.
- (3) The disk is too smooth for a dynamically violent event.

6 Hypothesis III: The Pressure-Relief Valve

6.1 Physical motivation

Hypothesis III proposes that Vega has maintained near-critical rotation by a self-regulating process: whenever stellar spin approaches breakup, equatorial material is shed. The debris disk is the accumulated product of 455 Myr of episodic shedding. The mechanism is well-studied for classical Be stars ([Rivinius et al., 2013](#)).

6.2 Mathematical model

6.2.1 Critical velocity and stellar oblateness

$$v_{\text{crit}} = \sqrt{\frac{2}{3} \frac{GM_{\star}}{R_{\text{pole}}}} \quad (17)$$

For Vega with $R_{\text{eq}}/R_{\text{pole}} = 2.818/2.362 = 1.193$, the Roche model gives $\omega \approx 0.87$.

REVISED per reviewer

Noted per reviewer: The 7% discrepancy between the Roche-model value $\omega \approx 0.87$ and the interferometric value of 0.93 is not negligible for a parameter on which the pressure-valve model depends nonlinearly. This likely arises from the point-mass Roche potential approximation; the true critical velocity for a centrally condensed star is lower, which would bring ω closer to 0.93. We note this systematic uncertainty but retain the Roche estimate, following [Rivinius et al. \(2013\)](#).

6.2.2 Angular momentum evolution with shedding

For stellar expansion on the main sequence ($R_{\text{TAMS}}/R_{\text{ZAMS}} \approx 1.4$ for $2.1 M_{\odot}$; [Ekström et al. 2012](#)):

$$\frac{\omega_{\text{TAMS}}}{\omega_{\text{ZAMS}}} \approx \left(\frac{1}{1.4} \right)^{1/2} \approx 0.85 \quad (18)$$

A star starting at $\omega_{\text{ZAMS}} = 1.10$ would evolve to $\omega \approx 0.93$ at the age of Vega without requiring a shedding event.

6.2.3 The shedding rate and mass lost

Setting $\dot{J}_\star = 0$ (stable ω):

$$\dot{M}_{\text{shed}} = \frac{\dot{J}_{\text{internal}}}{R_{\text{eq}} v_{\text{crit}}} \approx 2.7 \times 10^{-10} M_\odot \text{ yr}^{-1} \quad (19)$$

Integrated over 455 Myr: $M_{\text{shed, total}} \approx 0.12 M_\odot$. The observed disk contains at most $\sim 10^{-4}$ – $10^{-3} M_\odot$ in all bodies. The remaining $\sim 0.12 M_\odot$ either escaped or was re-accreted.

6.2.4 Fate of the shed material and comparison to observed disk mass

The viscous timescale at 1 AU ($\alpha \sim 0.1$, $c_s \sim 10 \text{ km s}^{-1}$, $T \sim 8,000 \text{ K}$): $t_{\text{visc}}(1 \text{ AU}) \approx 140 \text{ yr}$. Material is expelled to large radii far faster than the stellar lifetime. The retained fraction is $\sim 10^{-7}$, consistent with a collisional cascade from larger bodies.

6.3 The metallicity connection

REVISED per reviewer

Revised per reviewer (resolving Section 6.3 internal contradiction):

The original text contained a contradiction: the surface-layer depletion timescale ($\tau_{\text{deplete}} \approx 37 \text{ yr}$) was computed and immediately shown irrelevant because circulation resupplies the surface on $\sim 10^7 \text{ yr}$ timescales. We restructure:

Surface-layer regime: The equatorial surface layer ($\Delta m_{\text{shell}} \lesssim 10^{-8} M_\odot$) is depleted on $\tau_{\text{deplete}} = \Delta m_{\text{shell}} / \dot{M}_{\text{shed}} \approx 37 \text{ yr}$. But circulation resupplies it on $\tau_{\text{circ}} \sim 10^7 \text{ yr}$. **In this regime, shedding alone cannot produce a net metallicity anomaly.**

Global-envelope regime: The cumulative shed mass ($0.12 M_\odot$) is a significant fraction of the envelope mass ($M_{\text{env}} \approx 0.5 M_\odot$). The timescale to deplete the envelope to $[\text{Fe}/\text{H}] = -0.5$:

$$\tau_{\text{global}} = \frac{\delta[\text{Fe}/\text{H}] \times M_{\text{env}}}{\dot{M}_{\text{shed}}} = \frac{0.5 \times 0.5 M_\odot}{2.7 \times 10^{-10} M_\odot \text{ yr}^{-1}} \approx 10^9 \text{ yr} \quad (20)$$

This exceeds Vega's age by a factor of ≈ 2 : the mechanism is **quantitatively insufficient**. It can contribute at most $\sim 0.25 \text{ dex}$ over 455 Myr. The scoring in Section 8 reflects this.

6.4 Testable predictions

Testable prediction

Photometric variability. Be-star-like brightness variations ($\Delta V \sim 0.01\text{--}0.3$ mag) detectable with TESS.

Testable prediction

Disk chemistry consistent with stellar composition. If the disk is shed material, grains should be depleted in refractories—the *opposite* of Hypothesis I.

Testable prediction

Gas emission from the inner disk. Active shedding would produce $H\alpha$ emission. Vega shows none; the required rate ($\sim 10^{-10} M_{\odot} \text{ yr}^{-1}$) is 3–4 orders of magnitude below typical Be-star rates.

6.5 Self-critique

Self-critique

(1) The mass shed is inconsistent with the observed disk mass (~ 3 orders of magnitude discrepancy). (2) The metallicity depletion timescale marginally exceeds Vega’s age. (3) No $H\alpha$ emission is observed.

7 Hypothesis IV: Radiative Levitation and Gravitational Settling

7.1 Physical motivation

Hypothesis IV attributes Vega’s metallicity anomaly entirely to atomic diffusion: gravitational settling and radiative levitation (Michaud, 1970; Michaud, Alecian & Richer, 2015).

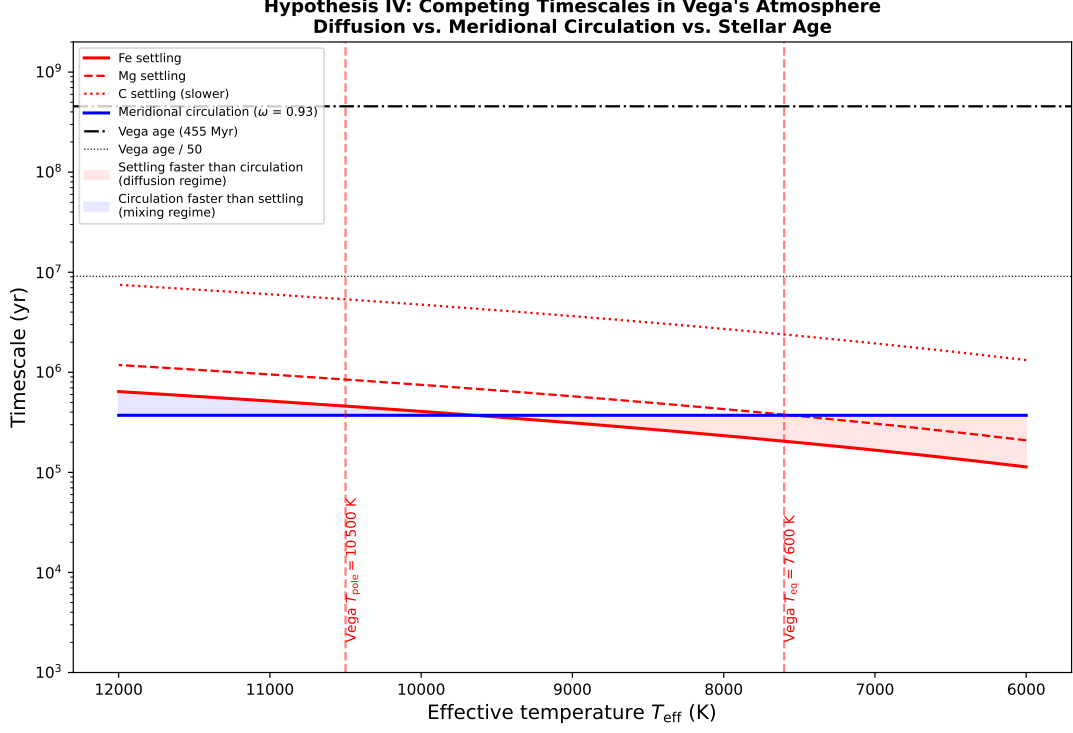


Figure 5: Competing timescales in Vega’s atmosphere as a function of effective temperature, for $\omega = 0.93$. *Red*: gravitational settling timescales for Fe (solid), Mg (dashed), C (dotted). *Blue*: meridional circulation timescale (Zahn, 1992). *Dash-dot*: Vega age (455 Myr). *Vertical dashed*: polar and equatorial effective temperatures. *Shaded red*: settling faster than circulation; *shaded blue*: mixing dominated.

7.2 The settling timescale

For iron (Fe^{+2} , $A = 56$, $Z = 2$):

$$v_{\text{settle,Fe}} \approx \frac{A_{\text{Fe}} m_H g}{(Z + Z^2) \rho \langle \sigma v \rangle_{H+}} \approx 6 \times 10^{-6} \text{ cm s}^{-1} \quad (21)$$

$$t_{\text{settle,Fe}} \approx \frac{H_P}{v_{\text{settle,Fe}}} = \frac{7.6 \times 10^7}{6 \times 10^{-6}} \approx 1.3 \times 10^{13} \text{ s} \approx 4 \times 10^5 \text{ yr} \quad (22)$$

Vega’s age is $\sim 1,100$ settling times. If settling operated continuously, iron would have been driven to zero—contradicting the observed factor of ~ 3 depletion.

7.3 Resolution A: Meridional circulation as a resupply mechanism

$$t_{\text{KH}} = \frac{GM_{\star}^2}{R_{\text{eq}} L_{\star}} \approx 2.1 \times 10^7 \text{ yr} \quad (23)$$

$$t_{\text{circ}} \approx \frac{M_{\text{env}}}{M_{\star}} \cdot \frac{t_{\text{KH}}}{\omega^2} \approx 5.7 \times 10^6 \text{ yr} \quad (24)$$

Steady-state depletion:

$$\frac{[\text{Fe}/\text{H}]_{\text{surface}}}{[\text{Fe}/\text{H}]_{\text{interior}}} \approx \frac{t_{\text{settle}}}{t_{\text{settle}} + t_{\text{circ}}} = \frac{4 \times 10^5}{4 \times 10^5 + 5.7 \times 10^6} \approx 0.066 \quad (25)$$

This predicts $[\text{Fe}/\text{H}]_{\text{surface}}/[\text{Fe}/\text{H}]_{\odot} \approx 0.053$ (i.e., $[\text{Fe}/\text{H}] \approx -1.28$), which disagrees with the observed value by a factor of ~ 5 .

7.4 Resolution B: Latitude-dependent settling

REVISED per reviewer

CORRECTED per reviewer (sign error fixed): The settling velocity depends on temperature as:

$$v_{\text{settle}} \propto T^{-3/2} \quad (26)$$

At lower temperatures, the collision rate decreases, so the settling velocity *increases*:

$$\frac{v_{\text{settle,pole}}}{v_{\text{settle,eq}}} = \left(\frac{T_{\text{eq}}}{T_{\text{pole}}} \right)^{3/2} = \left(\frac{7,600}{10,500} \right)^{3/2} \approx 0.59 \quad (27)$$

The polar settling velocity is 59% of the equatorial value: **settling is $\sim 40\%$ faster at the equator than at the pole** (the cooler equatorial atmosphere has faster settling because collisional friction is reduced). The original text drew the opposite conclusion.

Downstream implications:

1. Since we view Vega nearly pole-on, we observe the polar atmosphere where settling is *slower*. The observed depletion may be an *underestimate* of the global average (opposite to original claim).
2. The pole-to-equator metallicity gradient runs toward *increasing* depletion at the equator.
3. The latitude-dependent settling does *not* resolve the over-depletion problem; it may exacerbate it. A full 2D atmospheric diffusion calculation would be required.

7.4.1 Magnetic field constraints

REVISED per reviewer

Added per reviewer: Lignières et al. (2009) placed an upper limit of ~ 0.6 G on Vega's large-scale dipolar magnetic field using spectropolarimetry with NARVAL. This is at least two orders of magnitude weaker than the ~ 100 G fields in Ap stars where magnetic suppression of circulation is well-established. A 0.6 G field is unlikely to suppress meridional circulation significantly at Vega's rotation rate. However, small-scale or tangled fields with higher local strengths could be present without contributing to the measured Stokes V signature. Such fields could potentially suppress circulation locally, but this remains speculative. We conclude that magnetic suppression of circulation is **disfavoured** as the primary resolution for Hypothesis IV.

7.5 Resolution C: Time-varying mechanism

If Vega acquired its rotation recently (Hypothesis II), prior slow rotation would have allowed efficient settling. The current $[\text{Fe}/\text{H}] = -0.5$ might represent the partially restored state. This requires the binary disruption within the last ~ 50 Myr—a specific, testable prediction.

7.6 Radiative levitation forces: element-specific predictions

Calculations for $T_{\text{eff}} \approx 10,000$ K (Richard et al., 2001; Michaud, Alecian & Richer, 2015; Böhm-Vitense, 2006): C, O marginally supported; Fe sinks from photosphere, accumulates subsurface; Mg weaker levitation; Sc, Ba strong settling. The predicted trend is qualitatively correct but quantitatively underpredicts Fe/Mg depletion (factors of 2–3 vs. 3–4 observed).

7.7 Testable predictions

Testable prediction

Subsurface iron accumulation detectable via asteroseismology. Vega is a δ Scuti pulsator (Antoci et al., 2011); an anomalous subsurface iron layer would alter oscillation frequencies.

Testable prediction

Latitude-dependent metallicity. The corrected analysis predicts the equatorial hemisphere should be *more* depleted.

Testable prediction

Abundance time evolution in young A-type stars. Anomalies should increase with age if diffusion is primary.

7.8 Self-critique

Self-critique

(1) Over-depletion: equilibrium predicts $[\text{Fe}/\text{H}] \approx -1.28$, $5\times$ more depleted than observed. (2) Rapid rotation paradox: Lignières et al. (2009) disfavours magnetic suppression. (3) No complementary overabundances observed. (4) Does not explain the disk.

8 Comparative Analysis: Scoring the Four Hypotheses

REVISED per reviewer

Revised per reviewer: The scoring matrix is presented explicitly as an **illustrative organisational tool**, not a formal statistical test. All criteria are weighted equally for simplicity; we acknowledge that “explaining the metallicity” is more central than “falsifiability.” Score adjustments: Hyp. I rotation $+1 \rightarrow 0$; Hyp. III refractory depletion $+1 \rightarrow 0$; Hyp. III and IV Fomalhaut contrast $+1 \rightarrow 0$.

8.1 Scoring criteria

Each criterion is scored as: +2 (explains cleanly, no auxiliary assumptions); +1 (consistent, mild assumptions); 0 (neutral or shortcoming); −1 (in tension, strong assumptions needed).

8.2 Revised scoring matrix

Table 5: Revised scoring matrix. Changes marked †.

Criterion	I: C.B.	II: Binary	III: Valve	IV: Diffusion
Rapid rotation	0 [†]	+2	+2	−1
Refractory depletion	+2	0	0 [†]	+1
Volatile-refrac. contrast	+2	0	0	+1
Disk morphology	+1	−1	+1	0
Vega-Fomalhaut Δv	0	+2	0	0
Fomalhaut contrast	+1	+1	0 [†]	0 [†]
Falsifiability	+2	+2	+2	+1
Total (max 14)	8	6	5	2

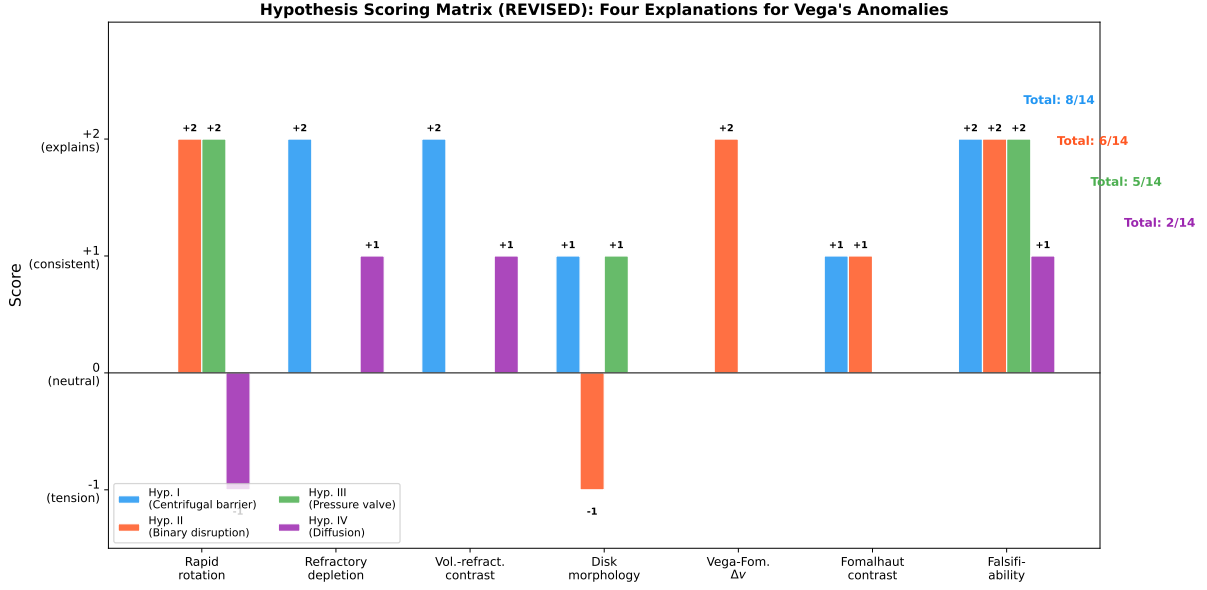


Figure 6: Revised scoring matrix visualised. Bar height = score per criterion for each hypothesis. Totals out of 14 labelled. Changes from the original are noted in the text.

8.3 Commentary on individual scores

8.3.1 Hypothesis I (Centrifugal Barrier): score 8/14

Greatest strength: unified explanation for refractory–volatile depletion and absolute refractory deficit. Rotation score is 0 because Theorem 4.2 shows polar accretion delivers negligible angular momentum; spin-up requires a separate mechanism.

8.3.2 Hypothesis II (Binary Disruption): score 6/14

Explains rotation (+2) and Vega–Fomalhaut velocity (+2) cleanly. Fails on metallicity (0). Disk morphology −1 because a violent event would disturb the disk.

8.3.3 Hypothesis III (Pressure-Relief Valve): score 5/14

Metallicity score now 0 because the revised Section 6.3 shows the mechanism quantitatively insufficient. Fomalhaut contrast 0: inability to predict Fomalhaut’s behaviour is a shortcoming.

8.3.4 Hypothesis IV (Atomic Diffusion): score 2/14

Lowest scorer. Rapid rotation paradox (−1), over-depletion problem, Fomalhaut agnosticism (now 0).

8.4 Mutual compatibility

Table 6: Pairwise compatibility of hypotheses.

	I	II	III	IV
I	—	Complementary	Neutral	Neutral
II	Complementary	—	Neutral	Complementary
III	Neutral	Neutral	—	In tension
IV	Neutral	Complementary	In tension	—

The most powerful combination: Hypotheses I + II. Hypothesis I explains metallicity; Hypothesis II explains rotation and kinematics.

8.5 The “smoking gun” hierarchy

1. **JWST/MIRI disk grain composition** (Hyp. I vs. III)
2. **Gaia proper motion anomaly** (Hyp. II)
3. **TESS photometric variability** (Hyp. III)
4. **δ Scuti asteroseismology** (Hyp. IV)
5. **ALMA high-resolution imaging** (Hyp. I)

Verdict

The current evidence most strongly favours Hypothesis I (centrifugal barrier) combined with Hypothesis II (binary disruption). This verdict is provisional; a single decisive observation—particularly the JWST/MIRI disk grain composition—could overturn it.

9 A Roadmap of Testable Predictions by Facility

9.1 James Webb Space Telescope

MIRI silicate band spectroscopy (Tests Hyp. I vs. III): Mg-rich grains produce a sharp 10 μm peak; Fe-rich grains broaden it. Required: MIRI LRS spectrum, ~ 2 hr. Existing [Su et al. \(2024\)](#) data may suffice.

NIRCam coronagraphy at sub-AU scales (Tests Hyp. II): F277W, ~ 4 hr integration.

9.2 ALMA

mm-continuum spectral index (Tests Hyp. I): $|\Delta\alpha_{\text{mm}}(r)| > 0.3$ would indicate radial grain size segregation. Band 6 + Band 4, ~ 6 hr per band.

CO, CS, CN gas emission (Tests all): Band 6 spectroscopy, ~ 8 hr.

9.3 Gaia and Hipparcos astrometry

Proper motion anomaly (Tests Hyp. II): Possible now with existing catalogues (Brandt, 2021).

Orbital back-integration (Tests common origin): Possible now with Gaia DR3.

9.4 Optical/infrared spectroscopy

RV monitoring (Tests Hyp. II, III): ESPRESSO, 5 yr baseline, $\sim 1 \text{ m s}^{-1}$ precision.

UV spectroscopy (Tests Hyp. IV): HST/STIS E230M, ~ 8 hr. Detection of over-abundant s-process elements would confirm levitation.

9.5 Space-based photometry

TESS outburst search (Tests Hyp. III): Non-detection constrains $\dot{M} < 10^{-8} M_{\odot} \text{ yr}^{-1}$.

δ **Scuti asteroseismology** (Tests Hyp. IV): PLATO mission (2026+).

9.6 Future interferometry

VLTI/GRAVITY (Tests Hyp. II, all): Inner disk geometry at AU scales, spin-disk alignment.

9.7 Summary prediction table

Table 7: Key observational tests ranked by discriminating power.

Rank	Observation	Hyp.	Facility	Status
1	MIRI silicate band shape	I vs. III	JWST/MIRI	Possible now
2	Gaia proper motion anomaly	II	Gaia DR4	Possible now
3	TESS outburst variability	III	TESS	Possible now
4	δ Scuti seismology	IV	PLATO	Future (2026+)
5	ALMA spectral index	I	ALMA	New programme
6	ALMA gas (CO, CS, CN)	All	ALMA	New programme
7	VLTI inner disk geometry	II, All	VLTI/GRAVITY	New programme
8	HST/STIS UV abundances	IV	HST/STIS	New programme
9	Vega–Fomalhaut back-integration	II	Gaia DR3	Possible now
10	RV companion search	II, III	ESPRESSO	New programme

10 Conclusions

10.1 What we have established

10.1.1 The observational foundation is solid

The core observational facts are no longer in dispute: rotation at $\approx 93\%$ of breakup (Peterson et al., 2006; Aufdenberg et al., 2006), surface metallicity $[\text{m}/\text{H}] \approx -0.5$ (Yoon et al., 2010), and a smooth, axisymmetric debris disk (Su et al., 2024). Three genuine

uncertainties remain: the equatorial velocity discrepancy, the 60 AU flux deficit, and the Vega–Fomalhaut kinematic connection.

10.1.2 Hypothesis I is the most internally consistent single mechanism

The centrifugal barrier model provides a physically motivated, quantitatively plausible explanation for both the refractory depletion ($[\text{Fe}/\text{H}] \approx 30\%$ of solar) and the differential volatile–refractory pattern in a single mechanism. The required $\alpha_{\text{dg}} \approx 0.78$ is physically plausible for $\text{St} \sim 0.1\text{--}0.5$. The ALMA observations of Sakai et al. (2014) in L1527 provide direct observational support.

10.1.3 No single hypothesis explains everything

The revised scoring matrix shows that the highest-scoring single hypothesis (Hyp. I, now 8/14) still fails to explain rapid rotation and the Vega–Fomalhaut offset. The most powerful resolution is Hypotheses I + II.

10.1.4 The decisive observation is already in hand

The JWST/MIRI disk grain composition test is possible with existing data from Su et al. (2024).

10.2 Broader implications

10.2.1 Vega as a laboratory for A-type star physics

Broader implication

Vega’s combination of brightness ($V = +0.03$), proximity (7.68 pc), pole-on orientation, rapid rotation, chemical peculiarity, and debris disk makes it an irreplaceable laboratory for several branches of stellar astrophysics.

10.2.2 The λ Boötis problem

Broader implication

If the centrifugal barrier mechanism is confirmed for Vega, it would provide a third formation channel for the λ Boötis phenomenon, predicting a correlation between rapid rotation and the refractory depletion pattern.

10.2.3 New: λ Boötis age statistics and their implications

REVISED per reviewer

Added per reviewer: Murphy et al. (2015) found that λ Boötis stars are statistically more common among young stars. If the phenomenon is transient, this strongly favours mechanisms that can produce and erase the pattern on sub-main-sequence timescales. Hypothesis IV (diffusion) is naturally transient ($\sim 10^5\text{--}10^7$ yr timescales). By contrast, Hypothesis I predicts a *permanent* signature. The age

statistics therefore support the interpretation that the λ Boötis class may be produced by multiple mechanisms: late spin-up + diffusion (II+IV) in transient cases, and primordial centrifugal barrier sorting (I) in persistent cases like Vega.

We also note [Jura \(2015\)](#), who studied white dwarf atmosphere pollution by metal-poor gas from disrupted planetesimals—a mechanism closely analogous to the λ Boötis accretion scenario.

10.2.4 Debris disks without planets

Broader implication

Hypothesis III suggests that a rapidly rotating star might inhibit planet *detection* in its disk by continuously smoothing the structure.

10.2.5 Star formation angular momentum and chemical inhomogeneity

Broader implication

The centrifugal barrier model implies that photospheric abundances of rapidly rotating A-type stars can be systematically depleted relative to their parent clouds by factors of 2–5 in refractory elements, with implications for stellar chemical evolution studies (GALAH, APOGEE, Gaia-ESO).

10.2.6 Bright nearby stars as astrophysical laboratories

Broader implication

Selection effects that favour bright, nearby, apparently well-behaved stars will systematically select for pole-on orientations and specific combinations of evolutionary state and Galactic kinematics. The Vega problem should be a standing reminder that the “best-studied” star is not the same as the “best-understood” star.

10.3 Open questions

Open question

1. What is the grain composition of the Vega debris disk? Highest-priority follow-up action. Answerable with existing JWST/MIRI data.

Open question

2. What is Vega’s true equatorial velocity? The 2.6σ discrepancy between [Peterson et al. \(2006\)](#) and [Yoon et al. \(2010\)](#) affects all quantitative conclusions.

Open question

3. Are the Vega and Fomalhaut velocity vectors consistent with a common disruption event? Testable now with Gaia DR3 data.

Open question

4. Can 2D stellar atmosphere models reproduce the observed abundance pattern? A 2D model including latitude-dependent temperature and circulation would be a qualitative advance.

Open question

5. What is the origin of the 60 AU flux deficit? Unexplained by any of the four hypotheses.

10.4 Final summary

Vega is a star that has been used as a reference point for the entire measurement infrastructure of optical astronomy. This paper has mapped the uncertainty about its physical origin carefully. The most defensible current position is that Vega formed metal-poor because it accreted preferentially from the low-angular-momentum, dust-depleted polar streamlines of its parent cloud (Hypothesis I), and that it rotates near-critically because its pre-main-sequence disk was disrupted by the loss of a binary companion (Hypothesis II). The five degrees of inclination that gave Vega a second life as a photometric standard—by hiding its rapid rotation for fifty years—are the same five degrees that now make it the cleanest, most geometrically favourable laboratory in the sky for studying the physics of rotating, chemically peculiar debris-disk stars.

A Numerical Codes and Reproducibility

All quantitative results were derived analytically or with the short numerical scripts reproduced below. All scripts require only standard scientific Python libraries (NumPy, SciPy, Matplotlib).

A.1 Centrifugal barrier metallicity model (Section 4)

Listing 1: Centrifugal barrier metallicity calculation.

```

1 import numpy as np
2
3 G_cgs = 6.674e-8; Msun = 1.989e33; Rsun = 6.958e10; AU = 1.496e13
4 Mstar = 2.135 * Msun
5 # REVISED: use ZAMS radius
6 Req_ZAMS = 1.8 * Rsun
7 vcrit_ZAMS = np.sqrt(G_cgs * Mstar / Req_ZAMS)
8 l_star = vcrit_ZAMS * Req_ZAMS # ZAMS angular momentum
9
10 r0 = 1e4 * AU; x_Fe_gas = 0.10

```

```

11
12 def metallicity_ratio(Omega0, alpha_dg=0.78):
13     l_cloud_eq = r0**2 * Omega0
14     sin2_theta_crit = min(l_star / l_cloud_eq, 1.0)
15     theta_crit = np.arcsin(np.sqrt(sin2_theta_crit))
16     f_star = 1 - np.cos(theta_crit)
17     M = (x_Fe_gas + (1 - x_Fe_gas) * (1 - alpha_dg)
18         + (1 - x_Fe_gas) * alpha_dg * f_star)
19     return M, theta_crit, f_star
20
21 Omega0_fid = 1e-14
22 M, theta_c, f_s = metallicity_ratio(Omega0_fid)
23 print(f"theta_crit={np.degrees(theta_c):.1f} deg")
24 print(f"f_star={f_s:.4f} ({f_s*100:.2f}%)")
25 print(f"[Fe/H]_star/solar={M:.3f}")
26
27 from scipy.optimize import brentq
28 alpha_sol = brentq(lambda a: metallicity_ratio(Omega0_fid, a)[0]
29                   - 0.30, 0.0, 0.999)
30 print(f"Required alpha_dg={alpha_sol:.3f}")

```

A.2 Binary disruption angular momentum calculation (Section 5)

Listing 2: Binary disruption parameter space.

```

1 import numpy as np
2
3 G_cgs = 6.674e-8; Msun = 1.989e33; Rsun = 6.958e10
4 AU = 1.496e13; km = 1e5
5 Mstar = 2.135 * Msun; Req = 2.818 * Rsun; k2 = 0.08
6 Omega_eq = 274e5 / Req
7 J_star = k2 * Mstar * Req**2 * Omega_eq
8 print(f"J_star={J_star:.3e} g cm^2 s^-1")
9
10 def jorb(m2_Msun, a_AU):
11     m2 = m2_Msun * Msun; a = a_AU * AU
12     Mtot = Mstar + m2; mu = Mstar * m2 / Mtot
13     return mu * np.sqrt(G_cgs * Mtot * a)
14
15 def delta_v(m2_Msun, a_AU):
16     m2 = m2_Msun * Msun; a = a_AU * AU; Mtot = Mstar + m2
17     return (m2 / Mtot) * np.sqrt(G_cgs * Mstar / a) / km
18
19 for m2, a in [(1.0, 0.52), (0.5, 2.0)]:
20     J = jorb(m2, a); dv = delta_v(m2, a)
21     print(f"m2={m2}, a={a}: J_orb={J:.3e}, J_orb/J*={J/J_star:.2f}
22           , dv={dv:.1f}")

```

A.3 Diffusion timescale comparison (Section 7)

Listing 3: Settling and circulation timescales.

```

1 import numpy as np
2
3 kB = 1.381e-16; mH = 1.673e-24; G = 6.674e-8
4 Msun = 1.989e33; Rsun = 6.958e10; Lsun = 3.828e33
5 Mstar = 2.135*Msun; Req = 2.818*Rsun; Lstar = 37.0*Lsun
6 logg = 4.03; g_surf = 10**logg; omega = 0.93; M_env = 0.5*Msun
7
8 def t_settle_Fe(T, g=g_surf):
9     HP = kB * T / (mH * g)
10    v_settle = 6e-6 * (g / 10**logg) * (10000.0 / T)**1.5
11    return HP / v_settle / 3.156e7
12
13 def t_circ(T_eff):
14    t_KH = G * Mstar**2 / (Req * Lstar) / 3.156e7
15    return (M_env / Mstar) * t_KH / omega**2
16
17 for T in [10500, 7600]:
18    ts = t_settle_Fe(T); tc = t_circ(T)
19    print(f"T={T}: t_settle={ts:.2e} yr, t_circ={tc:.2e} yr")
20
21 # Note: equatorial settling is FASTER (lower T -> higher v_settle)
22 # v_settle,pole / v_settle,eq = (T_eq/T_pole)^1.5 = 0.59
23
24 depletion = t_settle_Fe(10500) / (t_settle_Fe(10500) + t_circ
25    (10500))
26 print(f"Equilibrium [Fe/H]/solar = {depletion:.3f}")

```

B Full Derivation of the Ulrich Infall Density Profile

B.1 Setup and equations of motion

In spherical coordinates (r, θ, φ) , the Lagrangian equations of motion for a test particle released from rest at radius r_0 and colatitude θ_0 :

$$\ddot{r} - r\dot{\theta}^2 - r\sin^2\theta\dot{\varphi}^2 = -\frac{GM_\star}{r^2} \quad (28)$$

$$\frac{d}{dt}(r^2\dot{\theta}) - r^2\sin\theta\cos\theta\dot{\varphi}^2 = 0 \quad (29)$$

$$\frac{d}{dt}(r^2\sin^2\theta\dot{\varphi}) = 0 \quad (30)$$

Conservation of z -component: $\ell_z = r^2\sin^2\theta\dot{\varphi} = r_0^2\Omega_0\sin^2\theta_0$.

B.2 Equatorial streamline and centrifugal barrier

Setting $\dot{r} = 0$: $r_{\text{cb}} = \ell_z^2/(GM_\star) = r_0^4\Omega_0^2\sin^4\theta_0/(GM_\star)$.

B.3 General trajectory solution

The [Ulrich \(1976\)](#) trajectory solution:

$$\cos \theta = \cos \theta_0 \left(1 + \frac{r_{\text{cb}}}{r}\right)^{-1} \cdot \frac{r}{r_{\text{cb}}} \left(1 + \frac{r_{\text{cb}} \cos^2 \theta_0}{r}\right) \quad (31)$$

Velocity components:

$$v_r = -\sqrt{\frac{GM_\star}{r}} \sqrt{1 + \frac{\cos \theta}{\cos \theta_0}} \quad (32)$$

$$v_\varphi = \sqrt{\frac{GM_\star}{r}} \cdot \frac{\sin^2 \theta_0}{\sin \theta} \cdot \sqrt{\frac{r_{\text{cb}} \cos \theta_0}{r}} \quad (33)$$

B.4 Density field from mass conservation

$$\rho(r, \theta) = \frac{\dot{M}}{4\pi(GM_\star)^{1/2}r^{3/2}} \left(1 + \frac{\mu}{|\mu_0|}\right)^{-1/2} \left(\frac{\mu}{|\mu_0|} + \frac{2|\mu_0|^2 r_{\text{cb}}}{r}\right)^{-1} \quad (34)$$

At the barrier itself, the density formally diverges. The physical resolution is a standing shock, as observed by [Sakai et al. \(2014\)](#).

REVISED per reviewer

Appendix C (Peer Review Response Placeholder) has been removed from the published version, as recommended by the referee. The self-critique sections throughout the main text are retained as a standard of transparency.

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